STAT7008 Programming of Data Science

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Consultation Hour: Tuesday 15:00--16:00

Assessment Methods

- There are three pieces of assignment in this course
- Each accounts for 5% of your final grade for the course
- The assignments are more like engineering assignments which can help you have a deeper understanding of the course materials.
- The final group project accounts for 85% of your final grade for the course
- There is no written test or exam in this course.

Application of Python Programming on US Stock Market Investment

Group Project

- The maximum number of students in a group is four.
- Once you have a formed a group, you need to elect a leader, who will be responsible for submitting the deliverables of the project on behalf of your group.
- The group leader needs to fill in the Group Information Excel file and submit it to Moodle (under Project Section) before September 13.
- I will assign a number to each group.
- Email me on hcheung4@hku.hk if you need help to find teammates.

Group Project cont.

- This group project involves three investment exercises.
- Note that they are virtual investment exercises only, and you are not supposed to investment real money onto any stocks.
- The three exercises are daily, weekly and monthly investments.
- For each exercise, you are given 1 million virtual US dollars to start with.
- You are not allowed to borrow money or short-sell stocks.
- 1 million virtual US dollars is what you have for each investment exercise.

Group Project cont.

- Each group is given 3 million virtual US dollars in total for the final group project.
- You must invest all the money on some listed companies in the US stock market.
- That is, you are not allowed to hold cash.
- You are not allowed to invest in ETFs (Exchange-Traded Funds) or indices.
- For each exercise, you can invest in at most 5 companies.
- You can ignore the lot size, and you may purchase fractional shares to ensure your total investment equals 1 million.

Group Project cont.

- Your investment decisions need to be made based on the closing prices of stocks on November 6, which is a Friday.
- The group leader needs to submit the decisions for daily, weekly and monthly investments on or before November 8 (on Moodle under the Project Section).
- Return on daily investment will be calculated using the closing prices of the chosen stocks on November 9, which is a Monday.
- Return on weekly investment will be calculated using the closing prices of the chosen stocks on November 13, which is a Friday.

Group Project cont.

- Return on monthly investment will be calculated using the closing prices of the chosen stocks on December 4, which is also a Friday.
- Note that we will use the closing price, not the adjusted closing price.
- You should notify me if there has been a dividend payment (i.e., ex-dividend date) during the evaluation periods.
- Assume there is no transaction cost.
- The return on investment for the entire portfolio will be calculated by the marking tutor.

Project Assessment

- The written report accounts for 60% of the final grade for the group project.
- The Python code accounts for 5% of the final grade for the group project.
- The presentation video accounts for 20% of the final grade for the group project.
- The portfolio performance accounts for 15% of the final grade for the group project.

Written Report

- It must be at least 2 pages long.
- It must be written by humans.
- The group leader will submit the written report (in pdf format) and Python code with sample datasets (in a single zip folder) onto Moodle on or before December 6.
- The grades for the written report and computer code will be shared by all students in the group.

Presentation Video

- It must be at most 20 minutes long.
- Suppose you plan to present for 20 minutes and there are 4 students in the group, then each student should present for 5 minutes.
- You could record your presentation using the Zoom account given by hku. https://its.hku.hk/software/zoom/
- The group leader will submit the video (in mp4 format) onto Moodle on or before December 6.
- Each student in the group will be given an individual grade for the presentation.
- Each student should tell me your name and UID before you present.

Portfolio Performance

- The returns on daily, weekly and monthly investments will be calculated by Jiacheng, who is the marking tutor for the course.
- His email address is u3011679@connect.hku.hk
- The grade for portfolio performance will be shared by all students in the group.
- Each group's grade will be based on how their performance ranks against the rest of the class.
- Even if your group was ranked at the bottom, you will still be given some marks for this part.
- The performance ranking will be posted out onto Moodle.

Learning Outcomes

- Have some basic understanding of Python programming
- Able to bridge the gap between theoretical and practical knowledge
- Able to combine artificial intelligence with human intelligence